

SMITKUMAR JAYENDRAKUMAR VELANI

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Education

Northeastern University, Boston, MA

2026 – 2028

Master of Science in Data Science

GPA: 4.0/4.0

Adani Institute of Infrastructure Engineering (GTU)

2021 – 2025

B.E. Information Communication & Technology

CGPA: 8.24/10

Work Experience

Unified Mentor Pvt. Ltd. | Data Analyst Intern Certificate

Jan 2025 – Apr 2025

Stock Market Data Analysis System 🔗 Link

Python | Pandas | NumPy | Matplotlib | Seaborn | yfinance | Tkinter

- Integrated **Yahoo Finance API** to fetch real-time **OHLCV** data for **4+ stocks**, automating data cleaning, deduplication and validation using **Pandas** across **250+ daily records**.
- Engineered **11 technical indicators** from scratch (SMA, EMA, Bollinger Bands, RSI, MACD) using fully vectorised **NumPy** and **Pandas** with no external TA library.
- Built an interactive **Tkinter dashboard** with **20 clickable charts** and computed **Sharpe Ratio**, **Max Drawdown** and annualised return metrics for quantitative risk-return profiling.

IBM SkillsBuild | Data Analytics Intern (CSRBOX) Certificate

Jun 2024 – Aug 2024

Plant Health Prediction System 🔗 Link

Python | NumPy | Flask | TensorFlow | OpenCV | Keras

- Deployed a **MobileNetV2 Transfer Learning** CNN achieving **89% diagnostic accuracy** across **15 disease classes** and **3 plant species** (Pepper, Potato, Tomato) on **PlantVillage** dataset.
- Built **OpenCV** preprocessing pipeline (denoising, HSV-based lesion analysis, Canny edge detection) and implemented **two-phase training** with EarlyStopping and ReduceLROnPlateau callbacks.
- Deployed end-to-end **Flask web application** with drag-and-drop interface and real-time CNN inference across all disease classes.

Projects

Global Economic Intelligence Dashboard 🔗 Link

Full-Stack AI | Data Science Project

Python | Flask | D3.js | Scikit-Learn | Groq LLaMA | ARIMA

- Built a real-time economic intelligence platform for **190+ countries** integrating **World Bank API** with **ARIMA GDP forecasting**, **Isolation Forest** anomaly detection, and **VADER sentiment analysis** on live news headlines.
- Implemented **K-Means clustering** to identify economically similar countries and a **recession risk predictor** using GDP growth, inflation and unemployment; deployed interactive **D3.js choropleth map** with animated GDP bar chart race (2000-2024).
- Integrated **Groq LLaMA 3.3** for AI-generated country economic summaries and investor risk ratings; added **WebSocket** real-time updates, currency tracker, and PDF report export via **Flask-SocketIO**.

RAG PDF Chatbot 🔗 Link

NLP | LLM | RAG Project

Python | LangChain | FAISS | HuggingFace | Gemini API

- Built an end-to-end **RAG pipeline** using **LangChain** and **FAISS** vector database, processing PDF documents into **500-word overlapping chunks** and storing as **384-dim semantic vectors**.
- Implemented **semantic search** using **HuggingFace sentence-transformers** (all-MiniLM-L6-v2) to retrieve top-3 relevant chunks per query, achieving accurate answers even when user wording differs from document text.
- Integrated **Google Gemini 2.5 Flash API** with custom **prompt engineering** to generate context-grounded answers, tested on 15-page research documents and achieving correct answers on factual queries.

Loan Default Prediction with MLflow 🔗 Link

MLOps | ML Pipeline Project

Python | XGBoost | MLflow | Scikit-Learn | Pandas

- Trained 4 ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost) on **32,581 real loan records**, tracked all experiments using **MLflow**; XGBoost achieved best performance with **93.4% accuracy** and **AUC: 0.947**.
- Performed end-to-end data preprocessing including missing value imputation, outlier removal, label encoding and feature scaling on **Kaggle Credit Risk Dataset**.
- Identified **loan grade**, **home ownership** and **loan percent income** as top predictors via XGBoost feature importance and saved best model using **joblib** for deployment.

Technical Skills

Languages: Python, SQL, JavaScript, PHP, HTML, CSS, SwiftUI

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, OpenCV, Flask, LangChain, D3.js, Bootstrap

Tools & Databases: PostgreSQL, MySQL, MongoDB, FAISS, MLflow, Flask-SocketIO, Power BI, Git, GitHub, Jupyter, VS Code

Machine Learning: CNN, Transfer Learning, DQN, Reinforcement Learning, RAG, LLM Integration, K-Means Clustering, ARIMA Forecasting, Anomaly Detection, Semantic Search, Vector Embeddings